

BOOTSTRAP

Themen:

- Bootstrap Einführung & Setup
- Grid System & Breakpoints
- Komponenten (Navigation, Cards, Forms)
- Utility Classes
- Responsive Design mit Bootstrap
- Customization & Best Practices

Was ist Bootstrap?

Bootstrap - Das beliebteste CSS Framework

Bootstrap ist ein **Open-Source CSS Framework** für schnelle und responsive Webentwicklung.

Warum Bootstrap?

- **Schnelle Entwicklung:** Vorgefertigte Komponenten
- **Responsive Design:** Mobile-first Ansatz
- **Konsistenz:** Einheitliches Design-System
- **Große Community:** Viel Support und Dokumentation
- **Anpassbar:** Eigene Themes und Styles

Was ist Bootstrap?

Bootstrap 5 Features:

- Kein jQuery mehr benötigt
- Flexbox-basiertes Grid System
- CSS Custom Properties
- Verbesserte API
- Kleinere Dateigröße

Setup:

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
```

Bootstrap Grid System

Das 12-Spalten Grid System

Bootstrap verwendet ein **flexibles 12-Spalten Grid System** basierend auf CSS Flexbox.

Grid Struktur:

```
<div class="container">                                <!-- Container -->
  <div class="row">                                       <!-- Zeile -->
    <div class="col-md-4">                                <!-- Spalte -->
      Content 1
    </div>
    <div class="col-md-8">                                <!-- Spalte -->
      Content 2
    </div>
  </div>
</div>
```

Bootstrap Grid System

Container-Typen:

- `.container`: Feste max-width für verschiedene Breakpoints
- `.container-fluid`: 100% Breite auf allen Breakpoints
- `.container-{breakpoint}`: Fluid bis zum spezifischen Breakpoint

Spalten-Klassen:

- `.col`: Gleiche Breite
- `.col-{number}`: Spezifische Spaltenanzahl (1-12)
- `.col-{breakpoint}-{number}`: Responsive Spalten

Responsive Breakpoints

Bootstrap's Breakpoint System

Bootstrap definiert 5 **Breakpoints** für verschiedene Gerätegrößen:

Breakpoint	Class Infix	Dimensions
X-Small	<i>None</i>	<576px
Small	sm	≥576px
Medium	md	≥768px
Large	lg	≥992px
X-Large	xl	≥1200px
XX-Large	xxl	≥1400px

Responsive Grid Beispiel:

```
<div class="row">
  <div class="col-12 col-md-6 col-lg-4">
    <!-- Mobile: 12 Spalten, Tablet: 6 Spalten, Desktop: 4 Spalten -->
    Content 1
  </div>
  <div class="col-12 col-md-6 col-lg-8">
    <!-- Mobile: 12 Spalten, Tablet: 6 Spalten, Desktop: 8 Spalten -->
    Content 2
  </div>
</div>
```

Mobile-First Ansatz: Styles gelten ab dem Breakpoint aufwärts!

Komponenten - Navigation

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#">My Website</a>

    <!-- Toggler für Mobile -->
    <button class="navbar-toggler" type="button"
      data-bs-toggle="collapse" data-bs-target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>

    <!-- Navigation Links -->
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item">
          <a class="nav-link active" href="#">Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">About</a>
        </li>
        <li class="nav-item dropdown">
          <a class="nav-link dropdown-toggle" href="#" data-bs-toggle="dropdown">
            Services
          </a>
          <ul class="dropdown-menu">
            <li><a class="dropdown-item" href="#">Web Design</a></li>
            <li><a class="dropdown-item" href="#">Development</a></li>
          </ul>
        </li>
      </ul>
    </div>
  </div>
</nav>
```

Komponenten - Cards

Bootstrap Cards - Flexible Content Container

Cards sind **vielseitige Content-Container** für Text, Bilder, Links, etc.

Basic Card:

```
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">Quick example text to build on the card title.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>
</div>
```

Komponenten - Cards

Card Grid Layout:

```
<div class="row">
  <div class="col-md-4 mb-4">
    <div class="card h-100">
      <div class="card-body">
        <h5 class="card-title">Service 1</h5>
        <p class="card-text">Description...</p>
      </div>
    </div>
  </div>
  <div class="col-md-4 mb-4">
    <div class="card h-100">
      <div class="card-body">
        <h5 class="card-title">Service 2</h5>
        <p class="card-text">Description...</p>
      </div>
    </div>
  </div>
</div>
```

Card Variants: .card-header, .card-footer, .text-center, .bg-primary

Basic Form:

Komponenten - Forms

```
<form>
  <div class="mb-3">
    <label for="email" class="form-label">Email address</label>
    <input type="email" class="form-control" id="email" required>
    <div class="form-text">We'll never share your email.</div>
  </div>

  <div class="mb-3">
    <label for="password" class="form-label">Password</label>
    <input type="password" class="form-control" id="password" required>
  </div>

  <div class="mb-3 form-check">
    <input type="checkbox" class="form-check-input" id="remember">
    <label class="form-check-label" for="remember">Remember me</label>
  </div>

  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Input Groups:

```
<div class="input-group mb-3">
  <span class="input-group-text">@</span>
  <input type="text" class="form-control" placeholder="Username">
</div>
```

Form Validation: .is-valid, .is-invalid, .valid-feedback, .invalid-feedback

Komponenten - Buttons & Alerts

Buttons - Verschiedene Styles und Größen

```
<!-- Button Variants -->
<button class="btn btn-primary">Primary</button>
<button class="btn btn-secondary">Secondary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-outline-primary">Outline</button>

<!-- Button Sizes -->
<button class="btn btn-primary btn-lg">Large</button>
<button class="btn btn-primary">Default</button>
<button class="btn btn-primary btn-sm">Small</button>

<!-- Button Groups -->
<div class="btn-group">
  <button class="btn btn-outline-primary">Left</button>
  <button class="btn btn-outline-primary">Middle</button>
  <button class="btn btn-outline-primary">Right</button>
</div>
```

Alerts - Nachrichten anzeigen:

```
<div class="alert alert-success alert-dismissible fade show">
  <strong>Success!</strong> Your action was completed.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>

<div class="alert alert-warning">
  <strong>Warning!</strong> Please check your input.
</div>
```

Utility Classes

Bootstrap Utilities - Schnelle Styles ohne Custom CSS

Spacing (Margin & Padding):

```
<!-- Syntax: {property}{sides}-{size} -->
<div class="m-3">Margin auf allen Seiten</div>
<div class="mt-4">Margin top</div>
<div class="px-2">Padding left und right</div>
<div class="mb-5">Margin bottom</div>
```

Display & Flex:

```
<div class="d-flex justify-content-center align-items-center">
  Zentrierter Content
</div>
<div class="d-none d-md-block">Versteckt auf Mobile</div>
```

Colors & Backgrounds:

```
<p class="text-primary">Primary Text</p>
<div class="bg-success text-white p-3">Success Background</div>
```

Typography:

```
<h1 class="display-4">Large Headline</h1>
<p class="lead">Lead paragraph</p>
<small class="text-muted">Muted text</small>
```

Spacing Scale: 0, 1 (0.25rem), 2 (0.5rem), 3 (1rem), 4 (1.5rem), 5 (3rem)

Responsive Design Patterns

Pattern 2: Feature Cards

```
<div class="container my-5">
  <div class="row g-4">
    <div class="col-md-6 col-lg-4">
      <div class="text-center">
        <i class="bi bi-gear fs-1 text-primary"></i>
        <h4 class="mt-3">Feature 1</h4>
        <p>Description...</p>
      </div>
    </div>
    <!-- Weitere Features... -->
  </div>
</div>
```

JavaScript Komponenten

Modal Dialog:

```
<!-- Button trigger -->
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">
  Open Modal
</button>

<!-- Modal -->
<div class="modal fade" id="myModal">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Modal title</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        Modal content goes here...
      </div>
      <div class="modal-footer">
        <button type="button" class="btn btn-secondary" data-bs-dismiss="modal">
          Close
        </button>
        <button type="button" class="btn btn-primary">Save</button>
      </div>
    </div>
  </div>
</div>
```

Weitere JS Komponenten: - **Carousel:** Bildslider - **Collapse:** Ausklappbare Bereiche
- **Tabs:** Tab-Navigation - **Tooltips:** Hover-Informationen - **Popovers:** Click-Informationen

JavaScript Komponenten

Weitere JS Komponenten:

- **Carousel:** Bildslider
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- **Popovers:** Click-Informationen

Customization & Themes

1. CSS Custom Properties überschreiben:

```
:root {  
  --bs-primary: #your-color;  
  --bs-font-sans-serif: 'Your Font', sans-serif;  
}
```

2. SCSS Variables (bei Sass-Kompilierung):

```
// _custom.scss  
$primary: #your-brand-color;  
$font-family-sans-serif: 'Your Font', sans-serif;  
  
// Dann Bootstrap importieren  
@import "bootstrap/scss/bootstrap";
```

3. Eigene Utility Classes:

```
.text-brand { color: #your-color; }  
.bg-gradient-brand {  
  background: linear-gradient(45deg, #color1, #color2);  
}
```

4. Bootstrap Icons:

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.7.2/font/bootstrap-  
icons.css" rel="stylesheet">  
<i class="bi bi-heart-fill text-danger"></i>
```

Best Practices & Performance

Bootstrap effizient einsetzen

✅ **DO:** - Verwende das **Grid System** für Layout - Nutze **Utility Classes** für kleine Anpassungen - Implementiere **Mobile-First** responsive Design - Verwende **Semantic HTML** mit Bootstrap Classes - Teste auf **verschiedenen Geräten**

❌ **DON'T:** - Überschreibe nicht zu viele Bootstrap-Styles - Vergiss nicht die **Accessibility** - Lade nicht unnötige JS-Komponenten - Verwende keine veralteten Bootstrap-Versionen

Performance-Tipps:

```
<!-- Nur benötigte Komponenten laden -->
```

```
<link href="bootstrap.min.css" rel="stylesheet">
```

```
<!-- CSS vor Body-Ende -->
```

```
<script src="bootstrap.bundle.min.js"></script>
```

Customization Workflow: 1. Starte mit Standard Bootstrap 2. Definiere eigene Farben/Fonts 3. Erweitere mit Custom CSS 4. Teste responsive Verhalten

Accessibility: Bootstrap ist WCAG-konform, achte auf alt-Attribute und ARIA-Labels!

Ressourcen:

- Bootstrap Official Documentation:
<https://getbootstrap.com/>
- Bootstrap Examples:
<https://getbootstrap.com/docs/5.3/examples/>
- Bootstrap Icons:
<https://icons.getbootstrap.com/>
- Bootstrap Themes:
<https://themes.getbootstrap.com/>